



Roll No:

B.PHARM.

(SEM. II) THEORY EXAMINATION 2025-26

PHARMACEUTICAL ORGANIC CHEMISTRY I

TIME: 3 HRS

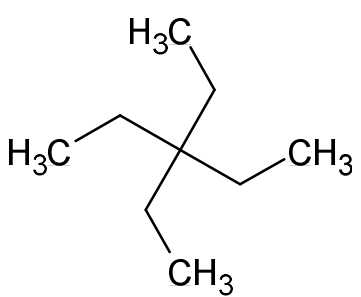
M.MARKS: 75

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

10 x 2 = 20

a.	Define structural isomerism with one suitable example.
b.	What is sp^2 hybridization? Explain with reference to alkenes.
c.	Write the IUPAC name for this compound: 
d.	Differentiate between electrophiles and nucleophiles.
e.	What is the order of reactivity of 1° , 2° and 3° alkyl halides in SN_1 reaction?
f.	Draw the structure of chloroform and write one medicinal use.
g.	Draw the structure of glycerol and mention one pharmaceutical use.
h.	What is crossed aldol condensation?
i.	How does the inductive effect influence the acidity of carboxylic acids?
j.	Draw the structure of amphetamine and mention one use.

SECTION B

2. Attempt any *two* parts of the following:

2 x 10 = 20

a.	Explain E_1 and E_2 reactions with suitable mechanisms. Compare them with respect to: i. Kinetics ii. Order of reactivity of alkyl halides iii. Factors affecting E_1 and E_2 reactions iv. Saytzeff's orientation
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b.	Differentiate between SN_1 and SN_2 reactions. Discuss their mechanisms, stereochemistry, carbocation rearrangement and factors affecting SN_1 and SN_2 reactions.
c.	Discuss nucleophilic addition reactions of carbonyl compounds in detail. Explain the following reactions with mechanism and synthetic importance: i. Aldol condensation ii. Cannizzaro reaction iii. Benzoin condensation

SECTION C**3. Attempt any five parts of the following:****7 x 5 = 35**

a.	Discuss electrophilic addition reactions of alkenes. Explain Markownikoff's and Anti-Markownikoff's orientation with suitable examples.
b.	Classify organic compounds with suitable examples. Explain chain isomerism, position isomerism and functional group isomerism with examples.
c.	Discuss the reactions of alkanes in detail.
d.	Describe the preparation and important reactions of alcohols. Explain any two qualitative tests used for alcohols.
e.	Write a note on i. Diels–Alder reaction ii. Ozonolysis of alkenes iii. Stability of conjugated dienes
f.	Explain the acidity of carboxylic acids and the effect of substituents on acidity. Mention one qualitative test used for carboxylic acids. Draw the structure of acetyl salicylic acid.
g.	Discuss the basicity of aliphatic amines and the effect of substituents on basicity. Explain one qualitative test used to distinguish primary, secondary and tertiary amines.